

ELEC5305 Project Proposal

Speech Enhancement in Noisy Environments Using STFT-Based Wiener Filtering

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1. Project Overview

Speech recordings made in real-world environments are often contaminated by background noise from sources such as traffic, air-conditioning, keyboards, and nearby conversations. This noise can reduce speech intelligibility and degrade the performance of applications such as online communication, voice recording, speech recognition, and hearing-assistance systems. The aim of this project is to design and evaluate a speech enhancement system that reduces additive background noise while preserving the important characteristics of the original speech signal.

The proposed solution will use Short-Time Fourier Transform (STFT) analysis together with Wiener filtering. The noisy speech signal will be divided into short overlapping frames and transformed into the time-frequency domain. A noise estimate will then be used to construct a Wiener filter that attenuates time-frequency components dominated by noise while retaining components more likely to contain speech. The enhanced signal will be reconstructed using the inverse STFT. The complete processing chain will be implemented and tested in MATLAB.

2. Background and Motivation

Speech enhancement is an important signal-processing problem because real speech recordings rarely occur in perfectly quiet conditions. Conventional noise-reduction methods improve speech quality by estimating the characteristics of the desired speech and the unwanted noise, then suppressing frequency components associated mainly with noise. Frequency-domain approaches are particularly useful because speech and noise can be analysed over short time intervals where their spectral characteristics are easier to separate.

Wiener filtering is a classical method for estimating a desired signal from a noisy observation. In a speech-enhancement application, the filter gain can be adjusted according to the estimated signal-to-noise ratio (SNR) in each time-frequency region. Previous research has shown that short-time spectral estimation and SNR-based filtering can provide effective noise reduction for speech signals. This topic was selected because it combines core concepts from ELEC5305, including sampling, Fourier analysis, STFT, spectral estimation, filtering, and objective performance evaluation, while remaining feasible to implement and test within one semester.

3. Proposed Methodology

The project will be developed mainly in MATLAB R2024a using audio input/output, spectral-analysis, filtering, and visualisation functions. Clean speech recordings will be obtained from a small open speech source or recorded locally. Controlled noisy test signals will be created by mixing the clean speech with selected background noises at several input SNR levels. This will make it possible to compare the enhanced signal with the original clean reference.

The first processing stage will perform STFT analysis. The signal will be divided into short overlapping frames using a suitable window, such as a Hamming window, and each frame will be transformed into the frequency domain. The noise power spectrum will be estimated from a noise-only segment or from the known noise signal used in the controlled tests. Based on the estimated speech and noise power spectra, a Wiener gain will be calculated for each time-frequency bin. Components with low estimated SNR will be attenuated more strongly, while components with higher estimated SNR will be preserved.

After filtering, the enhanced speech signal will be reconstructed using inverse STFT and overlap-add processing. The system will be tested with different noise conditions and input SNR values. Performance will be evaluated by comparing the noisy and enhanced signals using output SNR and SNR improvement. Waveforms and spectrograms will also be used to visually compare clean, noisy, and enhanced speech. If time permits, different STFT window lengths or noise-estimation settings will be compared to study their effect on performance.

4. Expected Outcomes

The main expected outcome is a working MATLAB prototype that accepts a noisy speech recording and produces an enhanced version with reduced background noise. The project is expected to demonstrate measurable SNR improvement in controlled test cases. Spectrogram comparisons should show reduced background-noise energy while preserving the main frequency components of speech.

The final project will include MATLAB source code, example audio files, waveform and spectrogram figures, performance measurements, the final report, and GitHub documentation. The public GitHub repository will contain the proposal, code, instructions for running the program, and example results so that the project can be reproduced and reviewed.

5. Timeline (Weeks 1-13)

Week	Task
Weeks 1-2	Finalise the project topic, define scope, and set up the GitHub repository.
Weeks 3-5	Review relevant literature; prepare speech and noise data; study STFT and Wiener-filtering theory.
Weeks 6-9	Implement the MATLAB processing pipeline: STFT, noise estimation, Wiener filtering, and signal reconstruction.
Weeks 10-11	Test several noise and SNR conditions, calculate performance metrics, and refine the implementation.
Weeks 12-13	Prepare figures and results; complete the final report and GitHub documentation.

6. References

1. Ephraim, Y., & Malah, D. (1984). Speech enhancement using a minimum mean-square error short-time spectral amplitude estimator. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 32(6), 1109-1121.
<https://doi.org/10.1109/TASSP.1984.1164453>
2. Scalart, P., & Filho, J. V. (1996). Speech enhancement based on a priori signal to noise estimation. *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2, 629-632.
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3. Plapous, C., Marro, C., & Scalart, P. (2006). Improved signal-to-noise ratio estimation for speech enhancement. *IEEE Transactions on Audio, Speech, and Language Processing*, 14(6), 2098-2108.
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